

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** CE

Commented [U1]: Please enter the name in Arial 12 font

2. **Subject Name:** Structural Modelling and Simulation Laboratory **L-T-P: 0-0-3**
Credits: 2

Commented [U2]: Please enter the details (use Arial 12 font)

3. **Pre-requisites:** 1) SOLID MECHANICS. 2) STRUCTURAL ANALYSIS

4. **Syllabus and reference books:**

Commented [U3]: 1. Please type the syllabus and reference books in the box provided below.

Syllabus: Introduction, basic modelling skills, quasi-static problems, eigen value buckling problems, problems with mesh constraints, problems with mesh independent constraints, dynamics problems, use of the adaptive meshing capability for problems with large deformation, problem involving soil-structure interaction, problem involving fluid-structure interaction, special applications - problems involving acoustics structural coupling, Problems involving thermo mechanical coupling

Please use "Arial" font with a font size of "12" and a line spacing of 1.0 for syllabus and reference books. Please use Justified Text (Ctrl+J) for the content entered in the boxes
Reference books should be numbered as 1), 2), 3)
Please provide all bibliographic details for the reference books

Reference Books:

- 1) Jacob, Fish, and Belytschko Ted. A first course in finite elements. Wiley, 2007.
- 2) Liu, Gui-Rong, and Siu Sin Quek. The finite element method: a practical course. Butterworth-Heinemann, 2013.

5. **Lecture-wise break-up:**

Commented [U4]: Please provide lecture-wise break-up in the table given below:

Sl. No.	Topic	No. of lectures
1.	Introduction: Importance of modelling and simulation in solving mechanics problems Choice of analysis procedures – dynamic vs. quasistatic, Idealizations of structures and systems, Analysis techniques, interpretation of results	3
2.	Analysis of building systems: Structures under gravity, wind and earthquake loads	9
3.	Quasi-static problems	3
4.	Eigen value buckling problems	3
5.	Problems with mesh constraints	3
6.	Problems with mesh independent constraints	3
7.	Dynamics problems	3
8.	Problem involving soil-structure interaction	6
9.	Problem involving fluid-structure interaction	6
Total number of hours		39

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